

Theorem 1 (Rolle's Theorem). Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and f is differentiable on (a, b) . Then if $f(a) = f(b)$, there is $c \in (a, b)$ such that $f'(c) = 0$.

1. Have you ever wondered why each assumption of Rolle's Theorem is necessary? No? Well, give an example of a function f which satisfies each of the following conditions, and yet $f'(c) \neq 0$ for all $c \in (a, b)$.
 - (a) A function f which is continuous on $[a, b]$ and differentiable on (a, b) (but we don't assume $f(a) = f(b)$).
 - (b) A function f which is continuous on $[a, b]$ and $f(a) = f(b)$ (but we don't assume f is differentiable on (a, b)).
 - (c) A function f which is differentiable on (a, b) and $f(a) = f(b)$ (but we don't assume f is continuous on $[a, b]$).

Theorem 2 (Mean Value Theorem). If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there is a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

2. Suppose that $f : [0, 3] \rightarrow \mathbb{R}$ is continuous on $[0, 3]$, differentiable on $(0, 3)$ and satisfies

$$f(0) = 1, \quad f(1) = 2 \quad \text{and} \quad f(3) = 2.$$

- (a) Show that there is some point $a \in (0, 3)$ such that $f(a) = a$.
- (b) Show that there is some point $b \in [0, 3]$ such that $f'(b) = \frac{1}{3}$.
- (c) Show that there is some point $c \in [0, 3]$ such that $f'(c) = \frac{1}{4}$.

Definition 3. A function f is called *Lipschitz continuous* if there exists some $C \in \mathbb{R}$ where

$$|f(x) - f(y)| \leq C \cdot |x - y|$$

for all x and y .

3. Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable, and assume f' is continuous on $[a, b]$. Prove that f is Lipschitz continuous on $[a, b]$.
4. Let I be an interval and $f : I \rightarrow \mathbb{R}$ be differentiable. Show that f is Lipschitz on I if and only if f' is bounded on I .